I2C communication

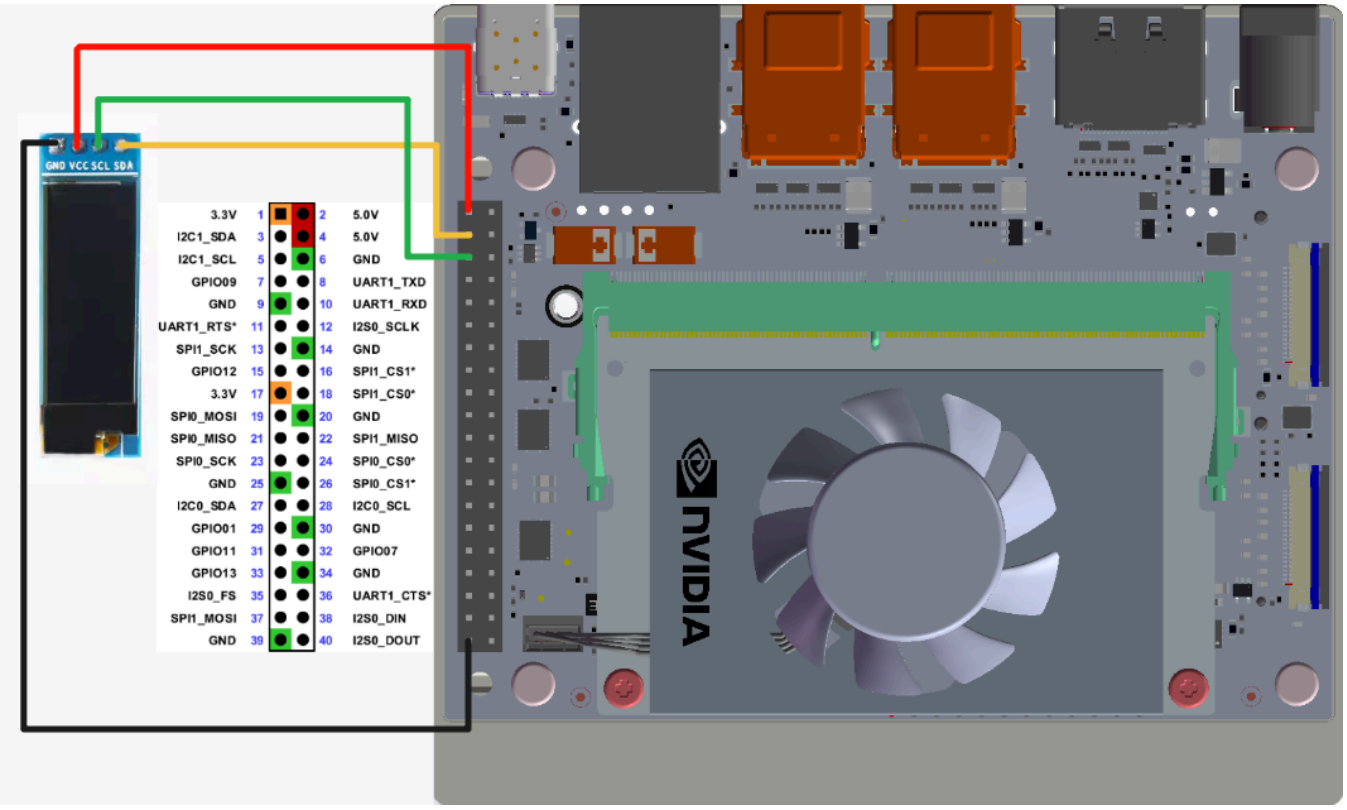
I2C communication

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1. GPIO pin diagram

Use a 0.91-inch OLED to test the I2C communication function, and connect it as follows:

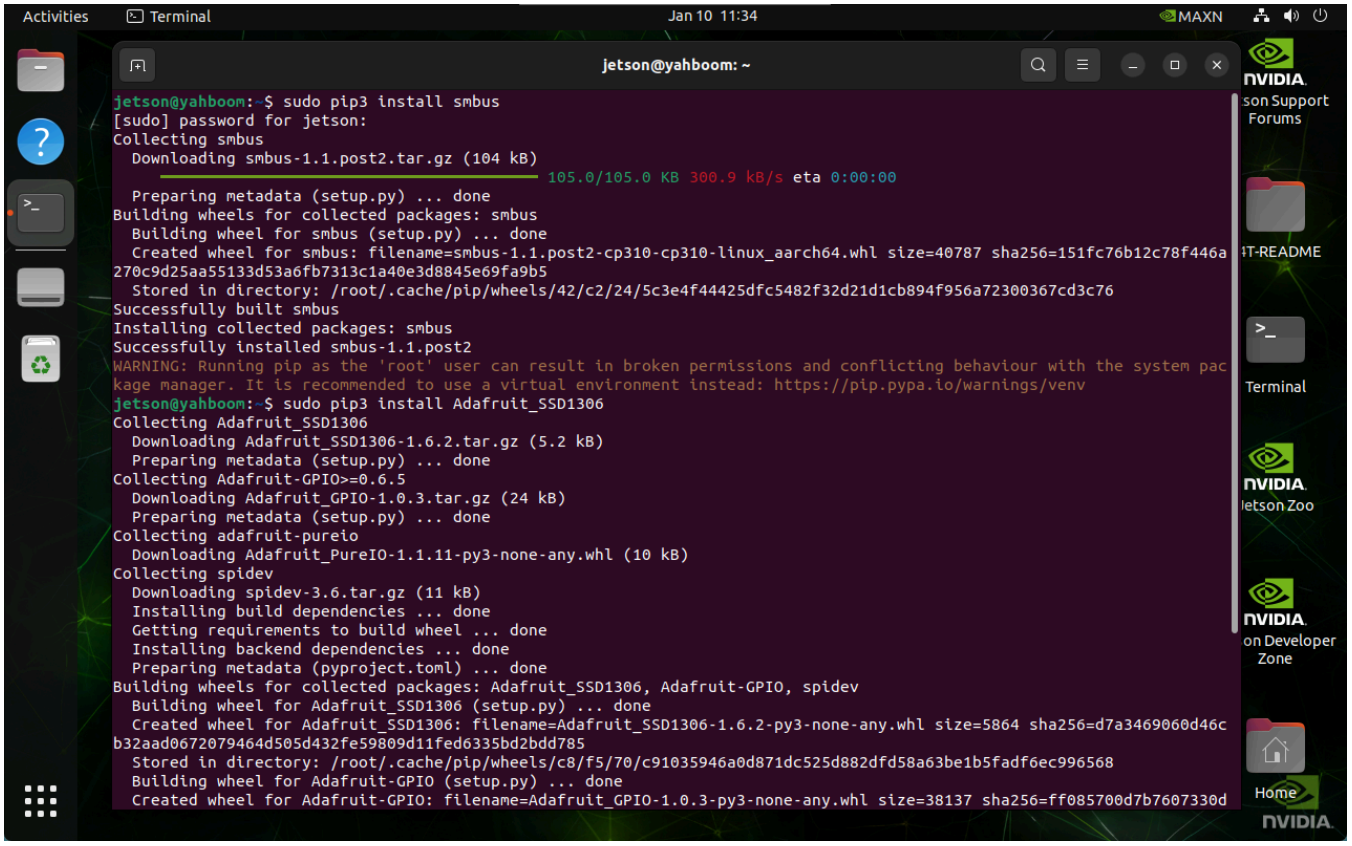
Note: Do not connect the pins incorrectly or cause a short circuit, as a mistake may damage the mainboard hardware!



2. I2C test

2.1. Install the dependencies

```
sudo pip3 install smbus --break-system-packages
sudo pip3 install Adafruit_SSD1306 --break-system-packages
```



2.2. I2C device

During normal development, we need to find the bus on which the I2C device is mounted and its device address.

2.2.1. Query the I2C buses

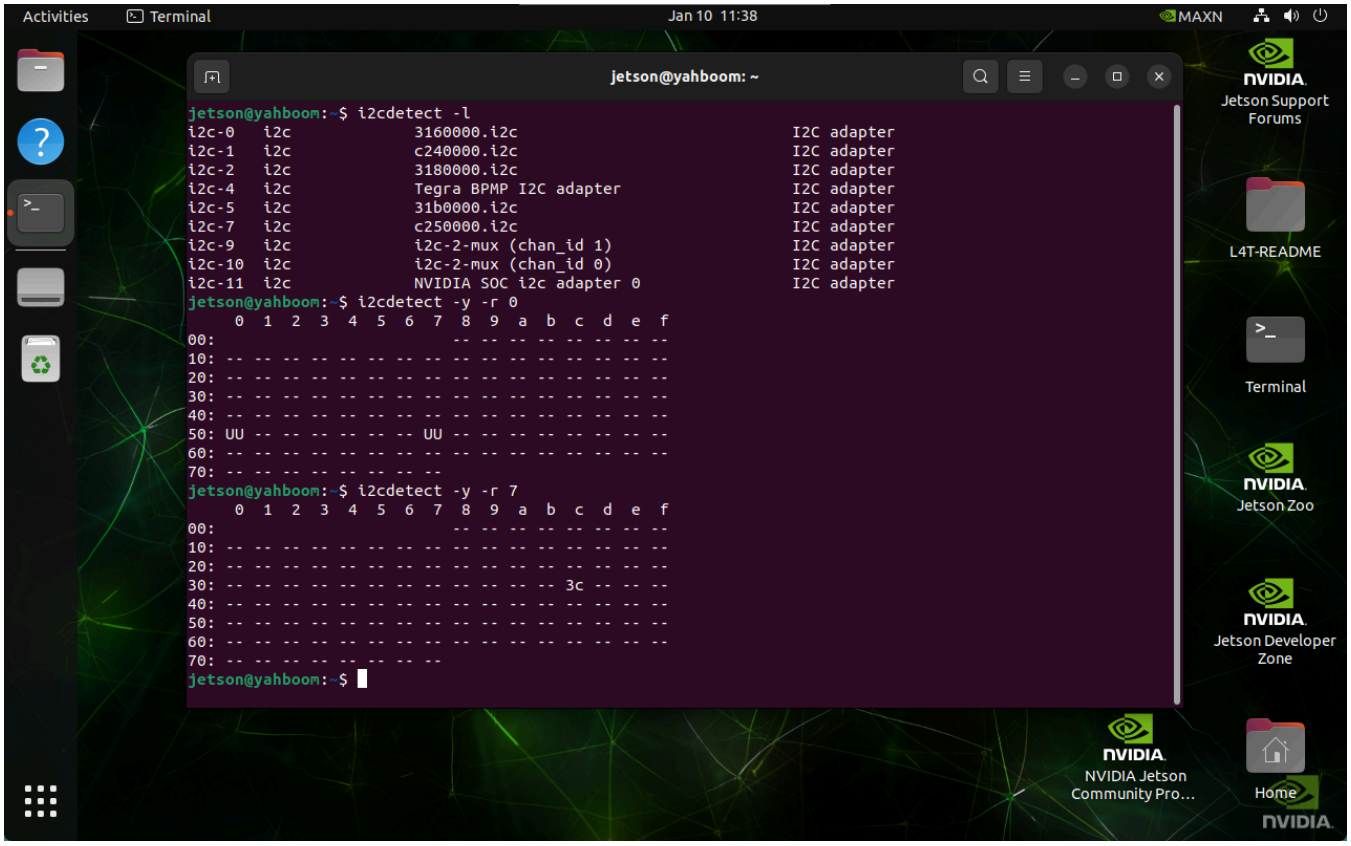
Enter the following command in the terminal to list all the buses of the device:

```
i2cdetect -l
```

2.2.2. Query the I2C devices

Enter the following command in the terminal to list the I2C devices on the specified bus: the I2C address of the oled is 0x3c

```
i2cdetect -y -r *
```



2.3. Run the program

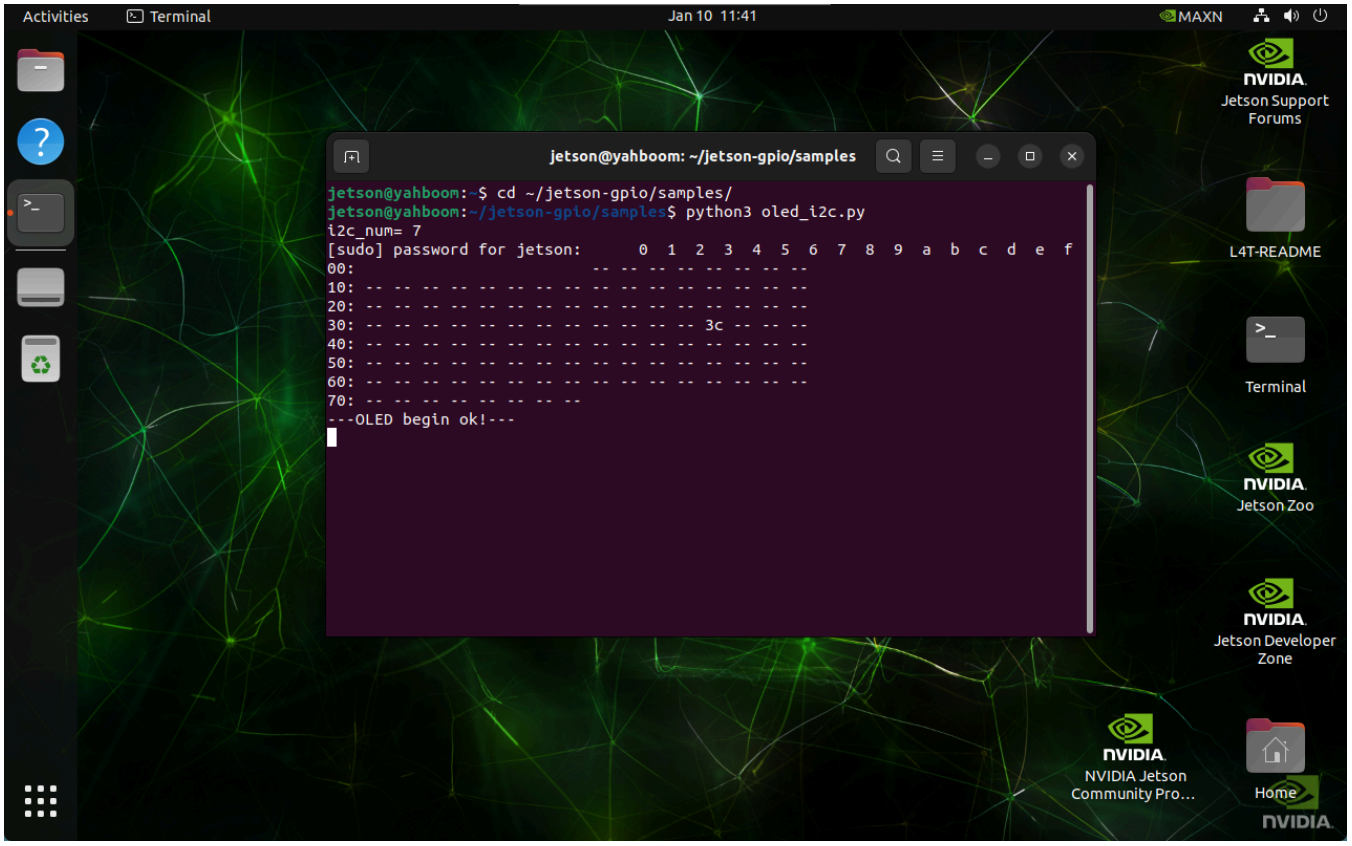
`oled_i2c.py` is not provided by the `jetson-gpio` library:

```
cd ~/jetson-gpio/samples/
```

```
python3 oled_i2c.py
```

3. Experiment effect

After the program is started, the OLED displays system information such as the system CPU usage, the system time and the memory usage:


Activities
Terminal
Jan 10 11:41
MAXN
jetson@yahboom: ~/jetson-gpio/samples
jetson@yahboom:~$ cd ~/jetson-gpio/samples/
jetson@yahboom:~/jetson-gpio/samples$ python3 oled_i2c.py
i2c_num= 7
[sudo] password for jetson:
0 1 2 3 4 5 6 7 8 9 a b c d e f
00: -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- 3c -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- --
---OLED begin ok!---
NVIDIA Jetson Support Forums
L4T-README
Terminal
NVIDIA Jetson Zoo
NVIDIA Jetson Developer Zone
NVIDIA NVIDIA Jetson Community Pro...
Home
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